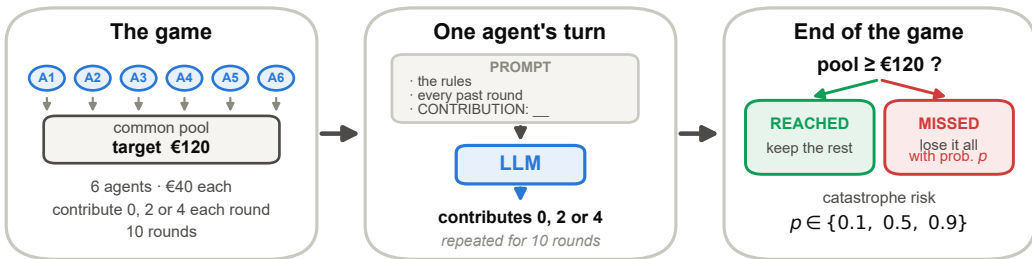


(a) A collective-risk dilemma, played by six LLM agents



(b) Two studies on the same games

Study 1 · Behaviour

Do the agents cooperate more when the risk is higher, as humans do (Milinski et al. 2008)?

full factorial · 7B to frontier

13 models × $p = .1 / .5 / .9$ × EN · VN

= 60 games per configuration

human benchmark: 0% · 10% · 50% reach

measures: target-reach rate · how much they pool

Study 2 · Comprehension

Is the deviation a real choice — or do they simply misunderstand the game?

the same prompt
+ one probe question

graded against the
engine's ground truth

Rules
the fixed rules

Time
read the history

State
sum the pool

20 questions · 3 axes · ground truth from the engine

measure: answer accuracy (does it understand?)